Observe, Imagine, Abstain: Detecting Hallucinated Geometry in Feed-Forward 3D Reconstruction

Anonymous CVPR submission

Abstract

*Feed-forward 3D foundation models reconstruct a scene from unposed photographs in a single pass. Where no photograph shows a surface, they fill the gap from learned priors, and they often assign the invented geometry a confidence indistinguishable from that of observed geometry. We call this geometric hallucination and define it operationally: a predicted point is hallucinated when it lies on a surface that no input view observes. HalluBench-3D produces such labels for free by hiding a 3D region of the ground-truth surface in every input view, and asks whether native confidence separates unsupported from supported pixels (*AUROC*) and how much unsupported geometry survives a threshold that keeps 90% of supported geometry, the Confident-Unsupported Rate (*CUR*). Across five models (VGGT, Pi3, Pi3X, MapAnything, Depth Anything 3), 18 DTU scans and 18 CO3Dv2 scenes, native confidence lets 24% to 81% of unsupported pixels through; an inpainted fill makes it worse, dropping views or adding a distractor changes nothing, variance over perturbed inputs is worse than native confidence, and the ranking of models changes between the lab rig and phone captures. We then ask what a detector of unobserved geometry can be. A small* support head *on the frozen multi-view features of the backbone predicts, per pixel, whether the output has multi-view support, and abstains in between. It needs no ground truth: its labels are the support count of the backbone's own prediction, the quantity a reprojection test measures. On four backbones and four datasets its training never touched (DTU, CO3Dv2 phone orbits, ETH3D building scans, Internet photographs of nine heritage landmarks) it separates single-view from supported geometry at* AUROC *0.93 to 0.99 against 0.55 to 0.89 for native confidence, lets 2 to 32 times fewer unsupported pixels through, matches a head trained on ground-truth visibility within 0.02, beats a per-image control that falls below chance off the lab rig, and beats reprojection consistency on three datasets and ties it on ETH3D at a fraction of the cost. A provenance variant, cross-view evidence attention over the same tokens, says which input views corroborate each pixel with no camera, at attribution* AUROC *0.93 to 0.96 on DTU (reprojection: 0.89 to 0.96) and below reprojection on building scans. Two cautions: a head trained on the benchmark's fills learns the fill, and no detector, this head included, fires on a filled patch that every view shows; five models of three families invent the same surface there, and what moves it is the fill itself. Code, benchmark, heads and the full run log will be released.*

1. Introduction

Feed-forward 3D reconstruction has moved from pairwise pointmaps [14, 31] to models that take dozens of unposed images and return dense geometry in one pass [13, 17, 29, 32]. They are trained to output geometry at every pixel, including pixels whose true surface no input view shows; the objective rewards a plausible guess, so they learn to guess, and the confidence map, the one signal a user has, does not say which parts were observed and which were guessed.

A conservator looking at a reconstruction of a dome sees a crack. Before sending someone up a scaffold, they want to know whether any photograph shows it or the model drew it from what domes usually look like; today the confidence map is bright in both cases.

We make three contributions.

1. **A definition and a benchmark.** We define hallucinated geometry as predicted points on surfaces that no input view shows. HalluBench-3D produces these labels by 3D-consistent masking of ground-truth surfaces, with controls on the same pixels under clean input. We report AUROC, CUR and CUR-ERR for five models on DTU and CO3Dv2, three fill types, an inpainted fill, and stress tests (view dropping, low parallax, a distractor image), each with cluster-bootstrap intervals and paired permutation tests.
2. **The backbone already knows, and a label-free way to read it.** Multi-view support is linearly decodable from the second half of the backbone's cross-view layers and absent from its per-image tokens; a linear probe gets within 0.015 of any head we trained. We read that signal with a small head whose labels come from the backbone's own prediction, so it trains on any capture with no ground truth, and abstains in between; a provenance vari-

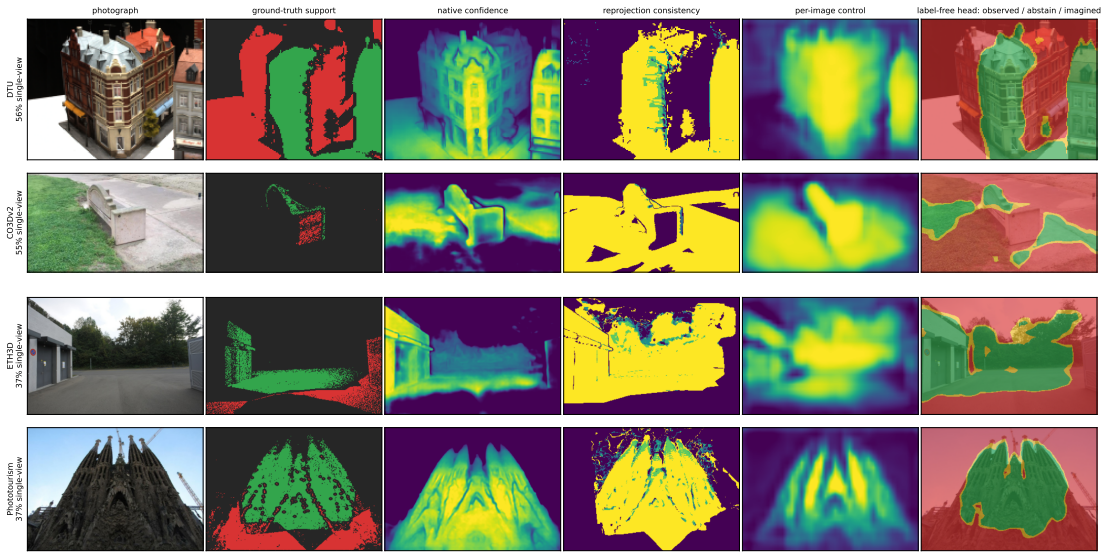


Figure 1. **What the label-free head says, on four captures it never saw.** One clean multi-view input per dataset (DTU rig, CO3Dv2 phone orbit, ETH3D DSLR, a tourist photograph of the Sagrada Família); ground-truth support (red: no other view sees the surface; green: two or more do; dark: no ground truth); VGGT's native confidence; reprojection consistency with its own cameras; the per-image control; the head's bands at the fixed thresholds (green observed, yellow abstain, red imagined). The head's labels came from the backbone's own predictions on DTU training scans; where native confidence lets half of the single-view pixels through a threshold that keeps 90% of the supported ones on DTU, the head lets 1.7% through. Run P6-dump.

ant reads *which* input views corroborate each pixel from the tokens alone, at the level of the model's own reprojection test on the rig and the phone orbits and below it on building scans. On four backbones and four datasets it never saw it passes two pre-registered tests against native confidence, a per-image control and a ground-truth-trained head (Section 7). Trained on the benchmark's synthetic fills instead, it learns the fill, a warning for masking-based supervision.

3. **Controls that a detector must pass, and one it fails.** Native confidence, perturbation variance, reprojection consistency, two hand-crafted cues, a linear probe and a head that sees one image at a time are scored on identical rows with paired tests. Put back to the benchmark's own filled patches, the head does not fire: it certifies agreement between views, which a consistent invention has.

Every number comes from a logged run with a run id; the pass conditions of Section 7 were written down, with dates, before the heads they judge were trained (supplement). The headline claim we test, not assume, is that native confidence fails to flag unsupported geometry across current models; it holds, more strongly on phone captures than on the lab rig.

2. Related work

Feed-forward 3D reconstruction. DUSt3R and MASt3R [14, 31] predict pointmaps for image pairs. VGGT [29], Pi3 and Pi3X [32], MapAnything [13], VGGT-Omega [30] and Depth Anything 3 [17] extend this to many unposed views in one pass. Each exposes a per-pixel confidence trained as a loss weight on observed data; none is trained to tell observed from unobserved surfaces. Pi3X replaces Pi3's near-binary confidence with continuous quality levels; MapAnything also offers a multi-view consistency confidence, a baseline below.

Hallucination in feed-forward 3D. Bratulic et al. [3] show that attention heads of DUSt3R, VGGT and DA3 keep matching correspondences after small patches are occluded, so the models hallucinate missing correspondences from learned priors; they measure attention and Sampson error on ShapeNet pairs, not confidence or 3D labels. Paul et al. [23] benchmark hallucination at the input level: unrelated scenes, repeated images and noise still produce dense geometry, scored by COLMAP-based consistency. 3D-Mirage [21] studies monocular depth under illusions; Li et al. [15] show that wide viewpoint spans produce unsupported geometry, and Dadajonov and Umarhojiyev [5] that unobserved parts of a room are not revealed by inference-time uncertainty; neither has per-point labels. FreeInpaint [22] fills masked regions on purpose, the mechanism these models apply unasked. Our definition is per point, with exact labels, and tests whether native confidence separates them.

Uncertainty for feed-forward 3D. ReVar3R [18] takes per-point variance over perturbed runs; it beats native con-

fidence on AUSE in most conditions but, by its authors' account, misses stable systematic bias, which a confidently filled region is. Trust3R [33] fits an evidential head on a frozen backbone, GeoCond [1] adds pose-level uncertainty, and Deng et al. [7] find the implied uncertainty of seven backbones at least 2.4× too small. All score uncertainty against ground-truth error, which conflates confident errors on observed surfaces with confident predictions on unseen surfaces; CUR isolates the second case. Closest to our detector, a linear probe on frozen VGGT features predicts *camera-pose* failure with AUROC 0.935 against 0.797 for native confidence [28], and a small head on the same features predicts pairwise co-visibility [34]; our support head is the dense, per-pixel analogue for unobserved *geometry*, trained on ground-truth visibility or on the backbone's own multi-view agreement with no label at all; the linear probe is one of our controls.

Generative and multi-sample 3D. Z3D [2] decodes hidden surfaces from frozen VGGT features, evidence that unobserved geometry is present in them. Feed-forward Gaussian splatting from unposed views [9] outputs pixel-aligned Gaussians with opacities, and ReconSplat [26] extrapolates beyond the observed views by design, all the more reason to say where it did. World models with a persistent Gaussian memory of the frames they generate [4] store imagined and observed surfaces alike, where a per-point support signal belongs. VolFill [20] finds sample variation 1.3× larger in occluded regions and diffusion depth ensembles [11, 12] give sample-disagreement uncertainty; a generative head over quantized geometry [6, 19] whose samples disagree where the model imagines is the natural successor to our discriminative head.

Hallucination in vision-language models. POPE [16] labels statements unsupported by the image; ours is the geometric analogue, geometry unsupported by any view.

3. Defining geometric hallucination

Given input views $I_1, \ldots, I_N$, a model returns for each view i a depth map D_i and a confidence map C_i. A pixel p in view i is *supported* if its ground-truth surface point is visible in at least one input view, and *unsupported* if it is visible in none. A model hallucinates at p when it outputs geometry at an unsupported p with high confidence.

Why confidence is the wrong tool. Native confidence is trained as a per-pixel loss weight on observed data: low where the regression target is noisy (edges, specular surfaces), high where it is smooth, so a region no view sees and the prior fills smoothly receives high confidence by construction; Section 5 measures the effect.

Metrics. All are computed per (model, scene, seed, radius, fill) and averaged with cluster-bootstrap intervals over scenes. Pixels whose confidence ties with τ count as kept, so a near-binary confidence (Pi3) is not favoured by the threshold.

- AUROC: $P(C_S > C_U)$ with ties counted half. 1.0 means every supported pixel outranks every unsupported one; 0.5 means confidence carries no information about support.
- CUR: choose τ so that 90% of supported pixels have $C \geq \tau$; CUR is the fraction of unsupported pixels with $C \geq \tau$. This is the hallucination rate you would experience after filtering by confidence.
- CUR-ERR: the fraction of unsupported pixels above τ whose AbsRel exceeds 0.1. Confident and wrong.
- AbsRel_U, AbsRel_S: depth error on unsupported and supported pixels after median scaling on supported pixels; controls: the same pixels under clean input (AbsRel_U ctrl) and the ratio of median confidence inside the hidden region to its clean value.

4. HalluBench-3D

Protocol. For each scene we take 10 views. A sphere is centred on a random ground-truth surface point in the reference view, with radius 0.10 or 0.18 of the object's bounding-box diagonal, resampled until it covers 2 to 35% of the reference view. Every pixel in every view whose ground-truth point lies inside the sphere is hidden (the protocol figure in the supplement). Hidden pixels are filled with gray, a blur of the original, uniform noise, or an inpainting of the surround [27]; the fill is an ablation so that results are not an artefact of a visible patch. Labels: U is hidden in all views; S is valid and outside a 6-pixel dilation of the hidden region; the ring in between is ignored. Five mask seeds per scene; masks are deterministic in (scene, seed, radius).

Datasets. *DTU* [8]: the 18 scans of the MVSNet validation split, 10 views spread over the 49-camera rig, model input 420 × 518. *CO3Dv2* [24]: 18 scenes, the two depth-carrying sequences of each of nine categories (apple, ball, bench, hydrant, mouse, teddybear, toytrain, toytruck, vase) from the single-sequence subset, with the official COLMAP depth as ground truth. The 10 views are spread over the full orbit, so the hidden region is seen from the back as well as the front; frames are 640 × 360 phone video, models run at 518 × 294 (or the transpose), and everything else, every threshold included, is unchanged from DTU. *ETH3D* [25]: nine DSLR building-scale scenes with laser-scanner ground truth on the undistorted pinhole frames (640 × 427, models 518 × 350), the reference frame and its nine most co-visible frames. *Phototourism* [10]: nine landmarks of the Image Matching Challenge 2020 set (listed in the supplement): Internet photographs with people and vehicles in

front of the monument, unordered viewpoints and focal lengths, COLMAP multi-view-stereo depth as ground truth; per landmark the best-connected photograph and its 19 most co-visible neighbours at 640×448; all nine pass the correctness guards (supplement). ETH3D and the landmarks are used only by the true-visibility protocol of Section 7, which draws its own view sets (3, 5 and 10 clean views, three rotations) on every dataset. Training-data overlap: VGGT, Pi3/Pi3X and Depth Anything 3 list CO3Dv2 among their training sets [17, 29, 32], MapAnything does not [13], and none lists DTU or ETH3D, so CO3Dv2 is in-distribution for three backbones and DTU and ETH3D out of distribution for all.

Models. VGGT-1B [29] (depth and depth confidence), Pi3 and Pi3X [32] (local point z and sigmoid confidence), MapAnything [13] (image-only metric mode), and Depth Anything 3 [17] (Large 1.1). Same views and resolution for all; fp16 on a T4. AnySplat [9], a feed-forward Gaussian-splatting model scored on each pixel's Gaussian depth with its opacity as confidence, is in the supplement from a partial run: the representation does not change the answer.

Statistics. Means carry 95% cluster-bootstrap intervals over scenes (10,000 resamples; each scene contributes its mean over seeds). Pairwise comparisons use a paired sign-flip permutation test on per-scene means (20,000 permutations), Holm-corrected within each family of pairs; the smallest p the test can return is 0.00005, written $p < 0.0001$ below.

5. Results on HalluBench-3D

5.1. DTU: native confidence leaks

Table 1 pools the full DTU configuration over the two natural-statistics fills, blur and noise (per-fill tables and the per-run spread in the supplement): CUR is 0.242 for VGGT, 0.281 for Pi3, 0.532 for Pi3X, 0.622 for MapAnything and 0.701 for Depth Anything 3, with AUROC 0.903, 0.912, 0.813, 0.743 and 0.711. Three tiers emerge. VGGT and Pi3 drop their confidence inside the hidden region (to 19 to 41% of its clean value) yet let one in four unsupported pixels through a threshold that keeps 90% of supported ones. Pi3X leaks one in two, and under a noise fill is as confident about geometry it cannot see as about geometry it can (ratio 1.09). MapAnything and Depth Anything 3 are close to blind: DA3's confidence inside the hidden region stays at 80 to 86% of its clean value for every fill, the gray patch included.

Pairwise, after Holm correction over the ten pairs: VGGT and Pi3 are indistinguishable ($p = 0.41$); both leak less than Pi3X ($p \leq 0.0015$), MapAnything and DA3 ($p = 0.0005$). Pi3X versus DA3 ($p = 0.052$), Pi3X versus MapAnything and MapAnything versus DA3 ($p = 0.19$) are not separated.

Confident and wrong. On DTU the imagined depth is accurate for four of the five models (ABSREL$_U$ 0.011 to 0.022, about 1.3 to 2× the clean value), so CUR-ERR stays near zero. The failure is that you cannot tell imagined from observed, not that the imagination is poor. MapAnything is the exception: 29% of its unsupported pixels are confident and wrong by more than 10% ABSREL, on top of a 10% error on supported pixels in image-only metric mode (the same at its native input size). The pass condition fixed before the benchmark was scaled (CUR above zero with intervals excluding zero, AUROC clearly below one on the natural fills) holds for all five models.

5.2. Stress tests: the leak is not a patch artefact

A reasonable objection to Table 1 is that a filled patch is visible and the model reacts to the patch, not to the absence of a surface. The stress table in the supplement answers it four ways. An inpainted fill, edge-continuous and without texture discontinuity, raises leakage further for VGGT ($+0.088$, $p = 0.031$) and Pi3X ($+0.068$, $p = 0.005$): Table 1 is a lower bound for a natural-looking occluder. Dropping from 10 to 3 views leaves CUR unchanged for VGGT and Pi3X ($p > 0.4$; DA3 moves at 5 views, $+0.11$, $p = 0.008$, and not at 3, which we read as noise): fewer views to cross-check does not make the models less sure, so native confidence follows single-view appearance, not multi-view agreement. With 3 views or low parallax, Pi3X's imagined depth degrades (ABSREL$_U$ 0.05, CUR-ERR up to 0.04) while its confidence does not move: the confident-and-wrong fraction, invisible on the easy rig, appears under harder capture. DA3 differs: a distractor image raises its leakage and low-parallax input halves it, consistent with a confidence that tracks appearance consistency across views rather than observation, which we flag rather than explain.

5.3. CO3Dv2: the ranking changes

Table 1 also gives CO3Dv2, the full protocol on phone captures with every threshold frozen from DTU. Pooled over blur and noise, CUR is 0.272 for Pi3, 0.570 for Pi3X, 0.651 for VGGT, 0.797 for DA3 and 0.810 for MapAnything; AUROC is 0.891, 0.732, 0.681, 0.620 and 0.576.

The ranking changes. Pi3 is the only model whose behaviour transfers from the lab rig to phone video (CUR 0.27 on both). The leak is larger on the dataset three of the models were trained on: seeing CO3Dv2 in training did not teach their confidence to notice a missing surface. VGGT, joint best on DTU, leaks 65% of unsupported pixels on CO3Dv2, 2.7× its DTU value, and under a noise fill is almost blind to non-observation (AUROC 0.59); MapAnything and DA3 let about four in five through. After

Table 1. **Native confidence leaks, on the lab rig and on phone captures.** Five models, full configuration (18 scenes × 5 mask seeds × 2 radii), pooled over the blur and noise fills; mean [95% cluster-bootstrap CI over scenes]. *conf ratio*: median confidence inside the hidden region over its clean value. Per-fill tables with every column in the supplement. Runs P1-full, P2-full, P2-co3d, P2-co3d-b; T4, fp16.

model	DTU, 18 scans				CO3Dv2, 18 scenes			
	AUROC	CUR	CUR-ERR	conf ratio	AUROC	CUR	CUR-ERR	conf ratio
VGGT-1B	0.903 [0.873, 0.929]	0.242 [0.181, 0.304]	0.000	0.22	0.681 [0.632, 0.729]	0.651 [0.566, 0.734]	0.031	0.70
Pi3	0.912 [0.881, 0.940]	0.281 [0.189, 0.379]	0.002	0.34	0.891 [0.845, 0.930]	0.272 [0.190, 0.360]	0.002	0.43
Pi3X	0.813 [0.762, 0.860]	0.532 [0.403, 0.663]	0.003	0.91	0.732 [0.680, 0.779]	0.570 [0.500, 0.641]	0.001	0.87
MapAnything	0.743 [0.687, 0.795]	0.622 [0.534, 0.707]	0.286	0.69	0.576 [0.521, 0.632]	0.810 [0.756, 0.863]	0.007	0.68
Depth Anything 3	0.711 [0.653, 0.763]	0.701 [0.614, 0.784]	0.002	0.83	0.620 [0.567, 0.670]	0.797 [0.728, 0.861]	0.000	0.88

Holm correction every pair is separated ($p \leq 0.010$) except VGGT versus Pi3X ($p = 0.23$) and MapAnything versus DA3 ($p = 0.73$). Across datasets, the intervals of VGGT, MapAnything and DA3 do not overlap; those of Pi3 and Pi3X do.

Imagination is accurate here too. ABSREL_U is within 0.005 of ABSREL_S for every model, so CUR-ERR stays below 0.04, and MapAnything's confident-and-wrong behaviour from DTU does not appear (ABSREL_S 0.041, CUR-ERR 0.007). These are everyday objects with strong shape priors; the models complete them well and report the completion as an observation.

Fills and scenes. On DTU every fill raised CUR over gray for VGGT, Pi3 and Pi3X. On CO3Dv2 noise raises VGGT's CUR by 0.20 ($p < 0.0001$) but blur does not ($p = 0.30$); Pi3X reacts to both (+0.26, +0.23), Pi3 weakly, MapAnything and DA3 not at all (p from 0.06 to 0.91): a confidence that does not change when the patch changes is not reading the patch. The leak is not driven by a few scenes (per-scene heatmap in the supplement); removing the one scene that missed the mask-transfer guard moves every pooled number by at most 0.013.

5.4. Cheap detectors do not help

Two obvious alternatives, scored on the DTU rows with blur, radius 0.18 and seed 0 (table in the supplement). Variance over $K = 4$ photometrically perturbed runs is far worse than native confidence on both backbones (AUROC 0.67 and 0.55 against 0.92 and 0.83, paired $p \leq 0.0007$) at four times the cost: the imagined geometry is stable under perturbation, the systematic bias ReVar3R's authors describe. Reprojection consistency with the model's own cameras is no better than chance (AUROC 0.49, 0.48): the imagined surface is cross-view coherent. MapAnything's multi-view consistency confidence leaks more than its learned confidence (AUROC 0.67, CUR 0.78 against 0.74 and 0.62 on the same rows, run P6-mapanything-mv).

6. Method: observe, imagine, abstain

The benchmark says that native confidence does not know what the model has seen. Rather than suppress the prior, which is what makes these models useful, we want the model to say which of its outputs came from the prior.

Architecture. A frozen backbone (VGGT, Pi3X, Pi3 or Depth Anything 3) provides geometry, native confidence, and the multi-view features that produced them: for VGGT and DA3 the tokens of the four cross-view layers their depth heads read (2048-d at 14×14 patches), for Pi3X and Pi3 the 2048-d decoder tokens feeding their confidence heads. The *support head* (LayerNorm and 1×1 projection per layer to 256 channels, sum, two 3×3 convolutions at patch resolution, one logit per patch upsampled bilinearly) outputs $P(\text{imagined})$; the benchmark scores $1 - P(\text{imagined})$. 3.3 M parameters on VGGT and DA3, 1.7 M on Pi3X and Pi3; the backbone is neither modified nor re-run.

Training labels. Three label sources, all on the 79 scans of the DTU training split, disjoint from the 18 benchmark scans and the 22 extension scans of Section 7; the views are clean, 3, 5 or 10 at random per step. *Self-support labels* (the main head, label-free): the backbone is run once; each pixel's *predicted* point is projected with the *predicted* depth and cameras into the other input views and counted as supported where it lands on their predicted surface (3% depth tolerance, the 20% least confident pixels dropped). A pixel no other view supports is the positive class, a pixel supported by two or more views outside a 3-pixel margin the negative class. This is the reprojection test of Section 5.4 used as a teacher, so the head must generalise beyond it to be worth having. *Visibility labels* (the upper bound): the same count from ground-truth depth and cameras. *Fill labels* (the cautionary variant): the benchmark's hidden sphere, placed on the predicted surface (gray, blur or noise; inpaint held out), with 15% clean samples. All three: class-balanced binary cross-entropy, 1500 AdamW steps (learning rate 10^{-3}, cosine, warm-up 100), one scene of random views per step, 10 minutes per backbone on one H200; hyperparameters fixed before any benchmark scan was evaluated.

Provenance head. The conservator also wants to know *which* photograph corroborates a pixel. The *provenance head* reads the same frozen tokens and returns, for every pixel of view i and every other input view j, the probability

p_{ij} that view j corroborates the surface there: after the per-layer projection and one 3×3 convolution, each target token queries the tokens of each other view with four-head attention (no cameras, no positional code), and a shared MLP over the attended evidence, the query and the peak attention weights returns the logit. The support score is the probability that no view corroborates the pixel, $1-\prod_{j\neq i}(1-p_{ij})$, so the head is scored on the cells and metrics of the other heads and the per-view probabilities are its added output. Labels are the per-view bits behind the support count (ground truth, or the backbone's own prediction for the label-free head), a class-balanced cross-entropy over the decided (view, pixel) pairs added to the support loss; optimiser, steps and data as for the other heads (3.1 M parameters on VGGT). Attribution is scored by *prov*-AUROC (does p_{ij} rank the views that see a point above those that do not, over every ordered view pair and decided pixel) and *top-1* (the highest-scoring view sees the point), against the reprojection test with the model's own cameras read per view; Gate 5 (supplement) fixed the pass conditions before any provenance head was trained.

Inference and compute. One backbone forward, one head forward; the head adds under 1% of the backbone's FLOPs. Three colours for the user: *observed* above $1-P(\text{imagined})=0.7$, *imagined* below 0.3, *abstain* between, fixed in advance; the metrics use the continuous score.

Controls and ablations. A head could learn an image cue (the patch; image borders and depth edges, where single-view pixels concentrate) rather than the missing support. The decisive control is the *per-image feature head*: the same head on per-image encoder tokens before any cross-view attention, same labels. Two hand-crafted scores test the same objection without training: the distance to the image border and minus the gradient of the predicted depth. Further controls: a *linear probe* on the same features [28] and on each cross-view layer alone, two extra seeds, and the held-out inpainted fill and CO3Dv2 for the fill-trained head.

7. Experiments on the method

Fill-trained head: what it learns. Its pre-registered test asked it to beat native confidence ($p<0.05$) and its per-image control on the held-out inpainted fill and on CO3Dv2. It scores AUROC 0.990 to 1.000 on every cell of the full configuration and the stress tests, and so does the control (within 0.003 of it on the inpainted fill and 0.02 on CO3Dv2; supplement); its formal clause passed on a 0.003 difference, a flaw of the criterion. Put to real single-view geometry (supplement), the same head scores *below* native confidence on every dataset: it learned the fill and nothing else, and the per-image control is mandatory for anyone training on masking labels.

What the tokens know. The per-image control falls below chance on CO3Dv2 and ETH3D, and so does the distance to the image border (indistinguishable from the control there, $p>0.3$): a head that cannot see across views learns the frame edge, which marks single-view pixels on the rig and not elsewhere. A linear probe on the same cross-view features scores within 0.015 of the convolutional head on every dataset: the information is in the features and any head is a reader. A probe on one VGGT layer at a time says where it appears (DTU / CO3Dv2): layers 4, 11, 17 and 23 of 24 score 0.87 / 0.48 (like the per-image control), 0.95 / 0.70, 0.99 / 0.92 and 0.99 / 0.88; support becomes linearly decodable in the second half of the aggregator.

Visibility head: the pre-registered test. Gate 3 was written before any visibility head was trained, with a margin this time: beat native confidence on AUROC and CUR ($p<0.05$) and the per-image control on AUROC by at least 0.02 ($p<0.05$), on DTU and on one unseen dataset. Table 2: the ground-truth-trained head reaches AUROC 0.990 to 0.993 on DTU against 0.77 to 0.84 for native confidence and 0.86 to 0.88 for the control (supplement table), $+0.11$ to $+0.13$ at every view count; off the rig it keeps 0.93 to 0.98 while the control falls *below chance* on CO3Dv2. Every paired difference is at the permutation floor ($p<0.0001$ on 18 scenes, 0.0043 on nine); Gate 3 passes on three datasets and three backbones.

Label-free head: the method. Gate 4, written before any label-free head was trained, asks the same of the head trained on the backbone's own support count, and adds: within 0.03 AUROC of the ground-truth-trained head on DTU, and ahead of reprojection consistency, its own teacher, on CO3Dv2 at $p<0.05$. It passes on all three backbones of Table 2, and on Depth Anything 3, the worst leaker of Table 1 (label-free head 0.98 / 0.88 / 0.97 / 0.93 on DTU / CO3Dv2 / ETH3D / landmarks against native 0.84 / 0.68 / 0.65 / 0.78 and control 0.76 / 0.44 / 0.75 / 0.73; supplement). The label-free head is within 0.02 of the ground-truth-trained head on every dataset, beats its per-image control by 0.10 to 0.12 on DTU and 0.35 to 0.50 elsewhere, and beats reprojection consistency by 0.04 to 0.07 on DTU and by 0.14, 0.05 and 0.04 on CO3Dv2 ($p<0.0001$, 0.006, 0.008); on ETH3D the two tie (±0.02, $p>0.15$): with adjacent DSLR views of a building, reprojection is as good as any head and the head's advantage is cost alone.

Tourist photographs, adaptation, more views. The landmark photographs are the capture the detector is meant for and the hardest for the alternatives: people stand between camera and monument and focal lengths differ from photo to photo, so the model's own cameras are least reliable and reprojection consistency scores 0.81 to 0.82. The label-free head scores 0.93 to 0.94 on all three backbones

Table 2. **True-visibility protocol: detecting single-view geometry from clean views.** Pooled AUROC and CUR over scenes and view counts (3, 5 and 10; 3 and 5 on CO3Dv2, where ten views around an object leave no single-view pixel). No head saw these scenes; the last row per backbone is the label-free head trained on the other half of the target dataset. Runs P4-vis*, P7-vis-imc, P6-adapt-*, P7-adapt-imc. DA3 and MapAnything native confidence on the same rows: AUROC 0.836 / 0.675 / 0.645 / 0.782 and 0.848 / 0.719 / 0.789 / 0.893. Intervals, per-view-count tables, every detector and paired tests: supplement.

detector	DTU		CO3Dv2		ETH3D		Photot.	
	AUROC	CUR	AUROC	CUR	AUROC	CUR	AUROC	CUR
VGGT-1B								
native confidence	0.772	0.530	0.754	0.723	0.715	0.566	0.802	0.451
reprojection consistency	0.954	0.082	0.793	0.693	0.975	0.069	0.824	0.340
support head, GT visibility labels	0.993	0.015	0.927	0.211	0.980	0.055	0.908	0.258
per-image control (self labels)	0.885	0.325	0.460	0.892	0.626	0.750	0.729	0.681
support head, label-free	0.992	0.017	0.931	0.199	0.980	0.051	0.930	0.191
label-free head, adapted on the target domain	–	–	0.841	0.474	–	–	0.926	0.203
Pi3X								
native confidence	0.820	0.493	0.697	0.846	0.887	0.304	0.839	0.450
reprojection consistency	0.914	0.164	0.885	0.224	0.977	0.054	0.814	0.361
support head, GT visibility labels	0.990	0.021	0.930	0.221	0.975	0.072	0.938	0.176
per-image control (self labels)	0.871	0.363	0.469	0.877	0.610	0.776	0.719	0.697
support head, label-free	0.988	0.025	0.939	0.190	0.963	0.103	0.940	0.166
label-free head, adapted on the target domain	–	–	0.964	0.094	–	–	0.929	0.198
Pi3								
native confidence	0.841	0.489	0.551	0.882	0.814	0.498	0.847	0.447
reprojection consistency	0.952	0.089	0.894	0.205	0.976	0.039	0.817	0.345
support head, GT visibility labels	0.992	0.016	0.934	0.203	0.953	0.135	0.932	0.201
per-image control (self labels)	0.888	0.320	0.471	0.895	0.654	0.750	0.724	0.683
support head, label-free	0.989	0.021	0.930	0.225	0.951	0.139	0.931	0.181

against 0.80 to 0.85 for native confidence (paired $p \leq 0.016$) and 0.72 to 0.73 for the per-image control; it beats reprojection consistency by 0.11 to 0.13 ($p = 0.0043$, the nine-scene floor). Trained on the target capture itself, still without labels (one half of a dataset, scored on the other), the head gains where the backbone's own geometry there is reliable, Pi3X on CO3Dv2 ($+0.025$, $p = 0.007$) and ETH3D ($+0.027$, $p = 0.008$), changes nothing on the landmarks (± 0.01, $p > 0.4$) and hurts VGGT on CO3Dv2 (-0.09, $p = 0.0004$), whose reprojection test, the teacher, scores 0.79 there against 0.89 for Pi3X; the DTU-trained head is the safe default.

What abstaining buys. CUR sets its threshold on the test scene itself, which a deployed filter cannot do. From the per-pixel scores (run P6-dump; supplement): the area under the single-view rate of the kept pixels against coverage is within 0.002 to 0.02 of the oracle for the label-free head on every dataset and two to four times larger for native confidence; keeping the half the head trusts most lowers the depth error of what remains by 15 to 44% (native confidence lowers it as much: it tracks error, the head tracks support). With the bands 0.3 and 0.7 fixed in advance, the head marks 57 to 87% of the labelled pixels *observed*, of which 0.5 to 5% are single-view (per-image control: 50 to 94% with 4 to 20%). Fusing the views with the backbone's own cameras and keeping the 80% of points the head trusts most raises the accuracy of the fused DTU cloud from 0.65 to 0.77 (native confidence 0.78, reprojection 0.76, control 0.70), while the oracle that drops single-view pixels first reaches 0.66: on DTU single-view geometry is nearly as accurate as supported geometry, so abstaining buys corroboration, not accuracy (ETH3D likewise; supplement). Multiplying the head by native confidence changes the head's AUROC by -0.055 to 0.004 over the sixteen backbone-dataset pairs and fuses to accuracy 0.79 at a single-view rate of 4% (native 0.78 at 5%): the product inherits the accuracy filter of native confidence and keeps most of the head's corroboration; neither substitutes for the other. A threshold set on the *other* scenes of a dataset lets 1 to 2% of single-view pixels through on DTU and 7 to 20% elsewhere, against 57 to 83% for native confidence; a DTU threshold applied unchanged elsewhere still works (supplement), whereas VGGT's native confidence has no transferable scale.

Which view saw it. Figure 2 and the supplement's provenance table score the provenance head on the same cells. Its support score matches the conv head within 0.01 AUROC on every cell but one (Pi3X on CO3Dv2, -0.025), so the structured output costs no detection. With no camera, the label-free head ranks the views that see a point above those that do not at prov-AUROC 0.93 to 0.96 on DTU, 0.83 to 0.88 on CO3Dv2, 0.82 to 0.84 on ETH3D and 0.80 to 0.85 on the landmarks (ground-truth-trained head: supplement); its top view sees the point for 98% of the supported pixels on DTU and 94% on CO3Dv2. Against reprojection read per view it is ahead on DTU for Pi3X, Pi3 and DA3 ($p \leq 0.0011$), -0.006 for VGGT, level on CO3Dv2 (-0.05 to $+0.05$) and behind on ETH3D (-0.16 to -0.09, $p = 0.004$) and the landmarks (-0.07 to -0.04), where the model's cameras and $N - 1$ projections per pixel know the frame geometry the tokens must infer. Gate 5 passes its support and top-1 clauses and fails its attribution clause; by its own fallback the conv head stays the method of record and the prove-

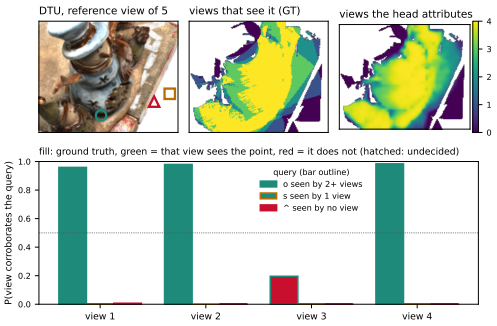


Figure 2. **Which view saw it** (VGGT, label-free provenance head, an unseen DTU scan, five views). Top: reference view with three query pixels, the number of other views that see each pixel (ground truth) and the number the head attributes. Bottom: the head's probability per other view for each query; bar fill is the ground truth. Run P10-prov.

nance head is reported as measured: per-view evidence at one forward and no camera, at the level of reprojection on the rig and the phone orbits, below it on building scans.

The benchmark's own question. The benchmark defines unsupported geometry as a surface no input view shows; the vis protocol's positive class is a surface only one view shows. Put back to the benchmark's filled patches (run P6-vis-on-fill), the visibility heads score AUROC 0.70 to 0.75 on DTU and 0.62 to 0.67 on CO3Dv2, the controls 0.60 to 0.65, border and depth edges 0.43 to 0.51, all below native confidence on the same rows. A head trained to find single-view geometry does not fire on a patch every view shows, for the reason the reprojection and perturbation baselines do not: the invented surface is coherent across views and features. It is coherent across models too: disagreement between two backbones on the same views (six pairs over five models, depths aligned by a label-free median ratio; run P8-xmodel) scores AUROC 0.52 to 0.61 on the DTU patches and 0.49 to 0.63 on CO3Dv2 (supplement); five models of three families fill the hole with the same surface. The detector answers "which pixels lack corroboration", not "which surfaces were invented". It is not, however, the same surface whichever pixels fill the hole. With the same hidden sphere filled three ways (gray, blur, noise) and the backbone run on each, the spread of the invented depth over the fills, aligned outside the hidden region, separates the patch from supported geometry at AUROC 0.82 to 0.98 on DTU and 0.62 to 0.96 on CO3Dv2 (run P9-refill; supplement), above native confidence for VGGT, Pi3X and DA3 and level for Pi3 on DTU, above it for VGGT, Pi3X and Pi3 and below it for DA3 on CO3Dv2; the median relative spread is 1.7 to 2.1% inside the hidden region against 0.1 to 0.6% outside, about the invented depth's own error, and refilling one view only moves the surface in that view and an order of magnitude less elsewhere (supplement). The invention is coherent across views, features and models because all of them read the same filled pixels; change the pixels and it moves. A real capture offers no second fill, so this is a diagnosis, not a detector, but it says where the invented surface comes from: the model reads the hole and draws what its pixels suggest.

8. Limitations and conclusion

The head certifies multi-view agreement, not truth: a surface every view agrees on passes, whether a camera saw it or the prior drew it, and the benchmark's filled patch is that case. The benchmark labels are a proxy for real occlusion, which the stress tests bound but do not remove. CO3Dv2 ground truth is COLMAP depth with 1 to 3% noise, and three backbones saw CO3Dv2 in training; the landmarks' ground truth is multi-view stereo with holes, so single-view labels there are a lower bound. MapAnything has only its native confidence on the vis protocol; static scenes only; all findings depend on the released checkpoints.

When a feed-forward 3D model has not seen a surface, it mostly does not say so: a quarter to four fifths of the geometry it invents passes a confidence filter tuned to keep 90% of what it observed. The answer is not to stop the models from imagining but to make them say when they do. The models already know: a linear probe on their cross-view features separates single-view from supported geometry at AUROC 0.99 where native confidence scores 0.8, and a small head trained on the backbone's own multi-view agreement, with no ground truth, reads the same signal on four backbones and four datasets it never saw, and a provenance head reads from the same tokens which views did the corroborating. What it cannot do sets the next problem: a surface the model invents consistently in every view passes every cross-view test we have, and five models invent the same one; what moves it is the pixels it was drawn from, which a real capture does not let us change.

LLM Usage Statement

A general-purpose LLM assistant was used, under the authors' direction, for parts of the code (the benchmark runner, the model wrappers, the run scripts and figure scripts), for literature lookup, and for editing the text of this paper, including grammar. The study design, the definitions and metrics, the choice and freezing of every hyperparameter, the execution and checking of every experiment, the verification of every reported number against its logged run, and the conclusions are the authors' own, and the authors are responsible for all content. The assistant was not used to generate results, figures from unlogged data, or references; every citation was checked against the published record. The tool is named in the camera-ready version.

References

[1] David Ahmedt-Aristizabal, Mohammad Ali Armin, Russell Tsuchida, and Lars Petersson. GeoCond: A Conditioning-Aware reliability adapter for Feed-Forward 3D reconstruction. *arXiv:2609.18465*, 2026. 3

[2] Denis M. Akola and David F. Fouhey. Zero-Shot novel depth synthesis using 3D foundation models scene representations. *arXiv:2609.04174*, 2026. 3

[3] Jelena Bratulić, Sudhanshu Mittal, Thomas Brox, and Christian Rupprecht. On geometric understanding and learned priors in Feed-forward 3D reconstruction models. *arXiv:2512.11508*, 2025. 2

[4] Jinhyeok Choi, JoungBin Lee, Jongmin Lee, Minkyeong Jeon, Jiyoung Kim, Kaeun Han, Yoonjin Yoon, and Seungryong Kim. L3G-World: Real-time infinite world generation with latent 3D gaussian memory. *arXiv preprint arXiv:2610.40614*, 2026. 3

[5] Alibek Dadajonov and Hikmatillo Umarhojiyev. Sharper frames, worse rooms: Viewpoint coverage dominates image quality in feed-forward 3D reconstruction. *arXiv preprint arXiv:2610.15854*, 2026. 2

[6] Haoge Deng, Ting Pan, Fan Zhang, Yang Liu, Zhuoyan Luo, Yufeng Cui, Wenxuan Wang, Chunhua Shen, Shiguang Shan, Zhaoxiang Zhang, and Xinlong Wang. Uniform discrete diffusion with metric path for video generation. *arXiv:2510.24717*, 2025. URSA. 3

[7] Nanxing Nick Deng, Qing Cheng, Niclas Zeller, and Daniel Cremers. A calibration audit of confidence in Feed-Forward 3D reconstruction models. *arXiv:2608.29705*, 2026. 3

[8] Rasmus Jensen, Anders Dahl, George Vogiatzis, Engin Tola, and Henrik Aanæs. Large scale multi-view stereopsis evaluation. In *CVPR*, 2014. 3

[9] Lihan Jiang, Yucheng Mao, Linning Xu, Tao Lu, Kerui Ren, Yichen Jin, Xudong Xu, Mulin Yu, Jiangmiao Pang, Feng Zhao, Dahua Lin, and Bo Dai. AnySplat: Feed-forward 3D gaussian splatting from unconstrained views. *arXiv preprint arXiv:2505.23716*, 2025. 3, 4

[10] Yuhe Jin, Dmytro Mishkin, Anastasiia Mishchuk, Jiří Matas, Pascal Fua, Kwang Moo Yi, and Eduard Trulls. Image matching across wide baselines: From paper to practice. *International Journal of Computer Vision*, 129:517–547, 2021. 3

[11] Jinyoung Jun, Lei Chu, Jiahao Li, Yan Lu, and Chang-Su Kim. Perfecting depth: Uncertainty-Aware enhancement of metric depth. *arXiv:2506.04612*, 2025. 3

[12] Bingxin Ke, Anton Obukhov, Shengyu Huang, Nando Metzger, Rodrigo Caye Daudt, and Konrad Schindler. Repurposing Diffusion-Based image generators for Monocular Depth estimation. In *CVPR*, 2024. 3

[13] Nikhil Keetha, Norman Müller, Johannes Schönberger, Lorenzo Porzi, Yuchen Zhang, Tobias Fischer, Arno Knapitsch, Duncan Zauss, Ethan Weber, Nelson Antunes, Jonathon Luiten, Manuel Lopez-Antequera, Samuel Rota Bulò, Christian Richardt, Deva Ramanan, Sebastian Scherer, and Peter Kontschieder. MapAnything: Universal Feed-Forward metric 3D reconstruction. *arXiv:2509.13414*, 2025. 1, 2, 4

[14] Vincent Leroy, Yohann Cabon, and Jerome Revaud. Grounding image matching in 3D with MASt3R. In *ECCV*, 2024. 1, 2

[15] Daisy Li, Kyle Gao, Quanyun Wu, Hanna Chomko, John S. Zelek, and Jonathan Li. When wider views fail: Stress-testing feed-forward 3D reconstruction. *arXiv preprint arXiv:2609.24839*, 2026. 2

[16] Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large Vision-Language models. In *EMNLP*, 2023. 3

[17] Haotong Lin, Sili Chen, Junhao Liew, Donny Y. Chen, Zhenyu Li, Guang Shi, Jiashi Feng, and Bingyi Kang. Depth Anything 3: Recovering the visual space from any views. *arXiv:2511.10647*, 2025. 1, 2, 4

[18] Sammam Mahdi, Fariha Binta Salim, Rakin Bin Rabbani, and Aniqua Nusrat Zereen. ReVar3R: Gauge-Aware perturbation uncertainty for Feed-Forward 3D reconstruction. *arXiv:2610.07883*, 2026. 2

[19] Fabian Mentzer, David Minnen, Eirikur Agustsson, and Michael Tschannen. Finite scalar quantization: VQ-VAE made simple. In *ICLR*, 2024. 3

[20] Tuan Duc Ngo, Chuang Gan, and Evangelos Kalogerakis. VolFill: Single-View amodal 3D scene reconstruction with volumetric flow matching. *arXiv:2605.31466*, 2026. 3

[21] Hoang Nguyen, Xiaohao Xu, and Xiaonan Huang. The 3D mirage: Probing and taming 3D hallucinations. *arXiv:2512.15423*, 2025. 2

[22] Jingyi Pan, Dan Xu, and Qiong Luo. Pose-free feed-forward 3D inpainting via learnable mask attention and support token refinement. *arXiv preprint arXiv:2610.11857*, 2026. 2

[23] Soumava Paul, Prakhar Kaushik, and Alan Yuille. Can these views be one scene? Evaluating multiview 3D consistency when 3D foundation models hallucinate. *arXiv:2605.18754*, 2026. 2

[24] Jeremy Reizenstein, Roman Shapovalov, Philipp Henzler, Luca Sbordone, Patrick Labatut, and David Novotny. Common objects in 3D: Large-Scale learning and evaluation of real-life 3D category reconstruction. In *ICCV*, 2021. 3

[25] Thomas Schöps, Johannes L. Schönberger, Silvano Galliani, Torsten Sattler, Konrad Schindler, Marc Pollefeys, and Andreas Geiger. A multi-view stereo benchmark with high-resolution images and multi-camera videos. In *CVPR*, 2017. 3

[26] Giuseppe Stracquadanio, Kevin Raj, Julia Grabinski, and Stefan Roth. ReconSplat: Generalizable 3D scene reconstruction beyond observed views. *arXiv preprint arXiv:2608.28895*, 2026. 3

[27] Alexandru Telea. An image inpainting technique based on the Fast Marching Method. *Journal of Graphics Tools*, 9(1), 2004. 3

[28] Adityan Thirumani Srinivasan and David Nordström. Does VGGT know what it does not know? *arXiv preprint arXiv:2610.35806*, 2026. 3, 6

[29] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. VGGT: Visual geometry grounded transformer. In *CVPR*, 2025. 1, 2, 4

[30] Jianyuan Wang, Minghao Chen, Shangzhan Zhang, Nikita Karaev, Johannes Schönberger, Patrick Labatut, Piotr Bojanowski, David Novotny, Andrea Vedaldi, and Christian Rupprecht. VGGT-ω. *arXiv:2605.15195*, 2026. 2

[31] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. DUSt3R: Geometric 3D vision made easy. In *CVPR*, 2024. 1, 2

[32] Yifan Wang, Jianjun Zhou, Haoyi Zhu, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Jiangmiao Pang, Chunhua Shen, and Tong He. π^3: Scalable Permutation-Equivariant visual geometry learning. *arXiv:2507.13347*, 2025. 1, 2, 4

[33] Zihao Zhu, Wenyuan Zhao, Nuo Chen, Chao Tian, and Zhiwen Fan. Trust it or not: Evidential uncertainty for Feed-Forward 3D reconstruction with Trust3R. *arXiv:2605.19539*, 2026. 3

[34] Filippo Ziliotto, Luciano Serafini, Lamberto Ballan, and Tommaso Campari. What VGGT knows about overlap: Probing geometric foundation models for co-visibility. *arXiv preprint arXiv:2607.09503*, 2026. 3